Sana Moin

AI/ML Engineer

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English (C1), German(B1)
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SKILLS AND TOOLS

- **Programming Languages:** Python, C, Java, JavaScript
- **Frameworks and tools:** PyTorch, TensorFlow, FastAPI, Flask, Celery, Qdrant, PostgreSQL
- **AI Domains:** Natural language Processing, Retrieval-Augmented Generation (RAG), speech recognition, using/fine-tuning multimodal models, machine learning, reinforcement learning, federated learning
- **Basics of Robotics:** Motion planning, sensor integration, robotic arm control, ROS
- **Cloud, Deployment & Tools:** Kubernetes, Docker, AWS, GitLab CI/CD
- **Collaboration:** Knowledge sharing, cross-functional teamwork

PROFESSIONAL EXPERIENCE

PINKTUM | Hamburg, Germany **Sep 2023 - Present**
AI/ML Expert (Engineer role)

- Developed a scalable asynchronous RAG application using FastAPI and Qdrant, deployed on AWS Cloud, enabling efficient personalized e-learning experiences for soft skills domain.
- Built an emotional and speech analysis system with Flask, Celery, Whisper, and LLMs, enhancing user engagement through advanced feature recognition.
- Restructured and stabilized backend systems, integrating Kubernetes for container orchestration and GitLab pipelines for CI/CD to ensure seamless deployment.

Fraunhofer CML | Hamburg, Germany **Jul 2021 - Aug 2023**
Research Assistant

- Designed and deployed MarFM, an automatic speech recognition(ASR) system for noisy maritime speech, achieving a 20% reduction in word error rate through transformer models, audio preprocessing, and synthetic data generation.
- Visited a maritime site in Finland to gather customer requirements and implemented distributed training using PyTorch Lightning and Distributed Data Processing (DDP), scaling models across GPUs and delivering near-real-time transcription with a high-performance ASR architecture.

Philips, Bengaluru, India **Jul 2018 - Oct 2020**
Software Engineer

- Built an Extraction-Transform-Load (ETL) pipeline for processing hospital data using JavaScript, Java, and SQL, achieving a 40% increase in processing speed. Collaborated with cross-functional teams, including DevOps, databases, and customer-facing teams, to deliver production-level upgrades and enhancements in a scrum environment.
- Developed a functional testing tool, reducing testing efforts by 80%. Independently resolved 60% of team bugs, earning multiple recognitions for ownership and fast delivery.

Philips Research, Bengaluru, India **Jan 2018 - Jun 2018**
Research Intern

- Performed ETL of medical text data and implemented a NER system using TensorFlow to automatically de-identify Protected Health Information in free-text medical records

EDUCATION

Universität Hamburg | Hamburg, Germany **Nov 2020 – Jan 2024**
MSc. Intelligent Adaptive Systems (AI and Robotics)

- **Thesis:** Created framework using PyTorch for training neural networks through federated learning with homomorphic encryption and quantization for efficient and private training on medical data.
- **Key Subjects:** Advanced computer vision, natural language processing, knowledge processing, data engineering, bio-inspired deep learning, robotics

Manipal Institute of Technology | Manipal, India **Jul 2014 – May 2018**
B.Tech. Computer Science and Engineering

- **Key Subjects:** Data structures and algorithms, operating system design, computer networks, distributed and cloud computing, data mining and information retrieval advanced database systems.

ACADEMIC PROJECTS

- **Robotic Navigation in Buildings:** Worked on a multi agent robotic navigation system using AI2-THOR, a simulation environment for reinforcement learning tasks, integrating CLIP embeddings for visual object identification and text-based contextual analysis to guide robots in indoor environments.

- **Offensive Meme Detection:** Built a system combining image processing and NLP models (BERT, bi-LSTM) to detect offensive content in memes, achieving high context understanding through multimodal analysis.

- **Hate Speech Detection in Movie Subtitles:** Designed a system using BERT and bi-LSTM to identify hate speech in movie subtitles, introducing a new dataset of 11,000 subtitles to improve detection accuracy.

- **Language Relationship Extraction:** Developed a model to extract meaningful relationships from texts in three less-common Indian languages using large language processing models (mBERT and RoBERTa), enhancing NLP capabilities for underrepresented languages.

PUBLICATIONS

- “How Hateful are Movies? A Study and Prediction on Movie Subtitles”-Proceedings of the 17th Conference on NLP, KONVENS 2021
- “De-Identification of Protected Health Information PHI from Free Text in Medical Records”-IJSPTM Vol 8, May 2019
- “The Performance Enhancement of Statistically Significant Bicluster Using Analysis of Variance”-Konkani A., Bera R., Paul S. (eds) Advances in Systems, Control and Automation. Lecture Notes in Electrical Engineering, vol 442. Springer, Singapore, 2017

BEYOND WORK

- Invited as a technical speaker in KI Summit Hamburg 2024 on the topic "Edge AI for Personalized Private Education Using Reinforcement Learning"
- Led three company-wide sessions on AI concepts, product architecture, and deployment workflows at PINKTUM.
- Received four recognitions at Philips for reflecting various Philips behaviour principles(‘Team up to Win’, ‘Eager to Improve & Inspire’ and two for ‘Take Ownership to deliver faster’)
- Advocate for ethical AI applications in healthcare and education to maximize societal impact
- Personal Interests: DIY art and redesign projects, solo travel, and photography.